Taha HOUMMADI

Software Engineering Student — ENSAM Meknes

AI & Intelligent Systems · Backend / Full-Stack Development

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Profile

Software engineering student at ENSAM Meknes (Intelligent Systems), building **AI-powered applications** with LLM/RAG, optimisation and full-stack development. **Capgemini Engineering intern:** developed an AI staffing platform (FastAPI, React, PostgreSQL, ChromaDB, OR-Tools). **Seeking a 3-month PFE from February 2027** in AI/ML or backend engineering.

Professional Experience

Software Engineer Intern — Capgemini Engineering *2026*

CasaNearshore, Casablanca

FastAPI · React.js · PostgreSQL · ChromaDB · OR-Tools · TensorFlow · LLM / RAG · JWT

- Architected **SkillsMatch**, an internal AI platform that recommends an optimal team for client projects: **FastAPI / React.js / PostgreSQL**, 12 REST routers and a 3-stage pipeline (role suggestion, candidate matching, team confirmation) shared by the web app and a conversational agent.
- Designed a hybrid CV pipeline (uploads, Gmail, LinkedIn): a local **Keras classifier** (OpenVINO/TensorFlow) categorises CVs before an **LLM** extracts skills and experience, limiting confidential data sent to external services.
- Implemented **semantic skill matching** with ChromaDB embeddings (exact match 1.0, near-match at cosine similarity ≥ 0.80 scored 0.5), avoiding brittle string matching such as "React.js" vs "ReactJS".
- Modelled team selection with **OR-Tools CP-SAT**: maximises candidate scores under hard budget and headcount constraints for a provably optimal team, with automatic budget relaxation reported when infeasible.
- Secured the platform with **JWT** (token revocation), **RBAC** and per-user **Fernet-encrypted** API keys; agent actions that modify data require explicit human approval.

Web Developer Intern — SRM Oujda *2025*

Société Régionale Multiservices, IT Department, Oujda

PHP Laravel · MySQL · JavaScript · HTML/CSS

- Developed a real-time web application to manage correspondence between company departments and external partners, with mail status tracking, email notifications, and an interactive interface.

Technical Skills

Languages

Python · C++ · Java · JavaScript · PHP · Dart

Backend & APIs

FastAPI · Node.js · Express.js · Laravel

REST · JWT/RBAC · OpenAPI

Frontend

React.js · HTML/CSS · TailwindCSS

AI / ML

LLM/RAG · TensorFlow · Scikit-learn

OpenCV · YOLOv8 · ChromaDB

Data & Optimization

PostgreSQL · MySQL · MongoDB

Pandas · OR-Tools

DevOps / IoT

Docker · Docker Compose · Git

ESP32 · Arduino · Raspberry Pi

Education

Software Engineering & Intelligent Systems

ENSAM Meknes (Engineering Degree) *2024 — Present*

Integrated Preparatory Program

ENSAM Meknes *2022 — 2024*

Baccalaureate, Mathematical Sciences A

Oued Eddahab High School, Oujda *2022*

Languages

Arabic: Native · **French:** B2 · **English:** B2

Academic Projects

Under-Train Inspection Robot — SIANA (Al Boraq High-Speed Rail) *2025 — Ongoing*

FastAPI · Docker Compose · React.js · IoT · Arduino · ESP32 · Raspberry Pi

- Designed and implemented the **containerised FastAPI backend** (Docker Compose) and **React.js dashboard** giving live visibility into robot sensor data, in a team project automating undercarriage inspection of Al Boraq trains.

AI Health Assistant — Medical Web Application *2025*

React.js · Node.js · PostgreSQL · LLM · RAG · Socket.io · JWT

- Built a **RAG-based medical assistant** and real-time consultation chat (Socket.io) on a secured microservices architecture, reaching **87% model accuracy** and **4.2/5 user satisfaction**.

MLOps — Facial Emotion Detection System *2025 — 2026*

Deep Learning · Python · MLOps · OpenCV · TensorFlow

- Built an end-to-end **MLOps pipeline** for an emotion-recognition model: training, model versioning and automated deployment, so every release is tracked and repeatable.

ENSAM Campus Apps — ReservENSAM & VoyageENSAM *2024 — 2025*

Node.js · React.js · JWT · MySQL

- **ReservENSAM:** implemented a booking system that removed scheduling conflicts and manual errors for student clubs' room reservations.
- **VoyageENSAM:** built a full-stack trip-management platform covering payments and refunds, protected personal data, and transport/accommodation, with scanned trip tickets that log attendance at each checkpoint (departure, Station A, return).

Competitions & Hackathons

RALLY-AI Future LAB — Ministry of Digital Transition (Merzouga) *Jun. 2026*

National hackathon: built a smart tourism mobile app (budget manager, holiday planner, automated booking) to help travellers discover Morocco's hidden places.

Agri-Food Hackathon — Ministry of Agriculture (AGRINOVA) *Apr. 2026*

National hackathon (Smart Farming, Smart Food): developed digital solutions for the agri-food sector.